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Professional Summary

Results-driven **Machine Learning Engineer** with **5+ years of experience** architecting and deploying production-grade AI solutions. Proven expertise in **deep learning**, **computer vision**, and **LLMs**. Track record of building end-to-end **ML systems** and **distributed data pipelines** that drive business impact. Passionate about solving complex challenges in **Computer Vision**, **Reinforcement Learning**, and **Large Language Models**.

Experience

AI Engineer (Freelancer) | *Remote* — *Jun 2025 – Jan 2026*

- Architected **real-time forecasting systems** for crypto and gold using **Binance API** and **WebSocket streams** for sub-second latency processing.
- Implemented **multi-horizon forecasting** with ensemble models (LSTM, GRU, Transformer, N-BEATS, TFT, Prophet, XGBoost), achieving ***45% RMSE reduction*** and **R^2 up to 0.97**.
- Engineered feature pipelines with **technical indicators (TA-Lib)** and rolling statistics using Pandas, NumPy, and scikit-learn.
- Optimized inference with **CUDA, ONNX, and TensorRT**, reducing latency by ***30%*** for live trading deployment.
- Built **real-time monitoring dashboards** using Weights & Biases, Matplotlib, and Plotly for drift detection and performance tracking.
- Implemented **automated retraining** with Optuna and MLflow, ensuring continuous model improvement.
- Designed **C++ optimization algorithms** for Maximum Clique Problem using Genetic Algorithms and RL. Created **Python bindings with PyBind11**, achieving ***38% improvement*** over baselines.
- Established **CI/CD pipelines** with GitHub Actions and Docker for reproducible builds and automated testing.

AI Engineer | *Rayan Pezhohan (Remote)* — *Nov 2023 – Jun 2025*

- Architected **real-time LLM platform** for real estate using **GPT-3.5/4, LangChain, Haystack**, and local models (LLaMA, Falcon) for structured queries and multi-use workflows.
- Implemented **RAG with FAISS and Milvus**, reducing hallucinations by ***45%*** and improving factual accuracy.
- Designed **low-latency inference pipelines** with optimized prompting, context management, and batching for concurrent request handling.
- Built **containerized backend** with Docker, Django, FastAPI, and PostgreSQL, exposing RESTful APIs for inference and session management.
- Developed **React frontend interfaces** for intuitive user interaction with LLM-powered features and real-time response visualization.
- Benchmarked cloud and local LLMs, implementing **model selection and hybrid routing** to optimize cost and latency.
- Applied **caching and API optimizations** with Redis and Celery, achieving ***30% latency reduction*** .
- Implemented **enterprise security** with JWT, OAuth2, and role-based access control for multi-tenant deployments.
- Trained **YOLOv8 models** for license plate detection (95% accuracy) and character recognition (92.5% accuracy) using PyTorch and OpenCV.
- Optimized training with **CUDA and mixed-precision**, reducing training time by ***60%*** and achieving **<160 ms inference per frame**.
- Built **high-performance backend** with Flask, asyncio, and socket programming, supporting **100+ concurrent users** with **<50 ms response times**.
- Used **MLflow** for experiment tracking and model versioning, ensuring full reproducibility.

Computer Vision Engineer | *Arooj (On-Site)* — *Dec 2021 – Nov 2023*

- Engineered **Hand Vein Detection Device** using **U-Net (89% IoU), SegNet, and YOLOv5** on **Raspberry Pi** with PyTorch, TensorFlow, ONNX, and TensorRT for edge inference.
- Developed **Shrimp Head Cutter System** using OpenCV and image processing, achieving ***>95% real-time accuracy*** in industrial settings.
- Reduced inference time by ***30%*** through **quantization, pruning, and multi-threaded optimization** while maintaining accuracy.

Research & Academic Experience

AI Researcher (Part-Time) | *Liverpool John Moores University (Remote)* *Jul 2024 – Present*

- Lead **deep learning research** in healthcare (infant cry analysis) and animal behaviour, specializing in **signal processing, feature extraction, and model optimisation**.
- Published in **IEEE Transactions on Affective Computing (Q1 Journal)** and currently submitting **7 additional Q1 papers**, demonstrating strong research impact.

AI & Python Mentor (Part-Time) | *Persian Gulf University (Remote)* *Aug 2025 – Present*

- Deliver instruction in **Python and AI concepts**, providing hands-on projects to build strong foundations for students and junior researchers.
- Mentor student teams in developing and deploying **AI applications and backend systems**, guiding projects from concept to production.

Education

Persian Gulf University **Bachelor of Electrical Engineering (Telecommunications)** | *2018 – 2023*

Projects

Car Dealing Platform

- Engineered **full-stack web app** using **React and Django** with vehicle listings, multi-parameter search, **JWT authentication**, and dealer dashboards for inventory management. Implemented **microservices architecture** for scalability.

Cat Voice Translator Mobile App

- Launched **cross-platform mobile app** using **React Native** leveraging **audio analysis and neural networks** to classify feline vocalizations into emotional states. Built end-to-end **data pipelines** for model training and deployment.

OCR

- Benchmarked **Vision Language Models** including **DeepSeekOCR, Ollama-OCR, QWEN2-VL, and PaddleOCR** for document processing. Designed **experimentation frameworks** to optimize accuracy and inference performance.

Tomato Leaf Disease Detection

- Developed **CNN-based classifier** using PyTorch and OpenCV to detect diseases in tomato plants, enabling **precision agriculture** through computer vision. Built **production ML systems** for real-time field deployment.

Publications

Infant Cry Analysis: A Survey of Datasets, Features, and Machine Learning Techniques
| *IEEE Transactions on Affective Computing (Q1 Journal)* *2025*

- Published in **IEEE Transactions on Affective Computing (Q1 Journal)**. Currently submitting **7 additional Q1 papers** in healthcare, animal behaviour, and intelligent systems.

Technical Skills

Languages : Python, C, C++, JavaScript, MATLAB
Machine Learning : PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers, LangChain, Haystack, Optuna, MLflow
MLOps & Production : MLflow, CI/CD, Model Deployment, Model Monitoring, Docker, Kubernetes, AWS, Distributed Systems, Microservices Architecture
Computer Vision : OpenCV, YOLO, U-Net, SegNet, Albumentations, ONNX, TensorRT
Backend & Data Pipelines : FastAPI, Flask, Django, PostgreSQL, Redis, MongoDB, RabbitMQ, Celery, Data Pipelines, Experimentation Frameworks
Frontend : React, React Native
Specialized : LLMs, RAG, Vector Databases (FAISS, Milvus), Time Series Forecasting, Reinforcement Learning, Edge AI, Production ML Systems